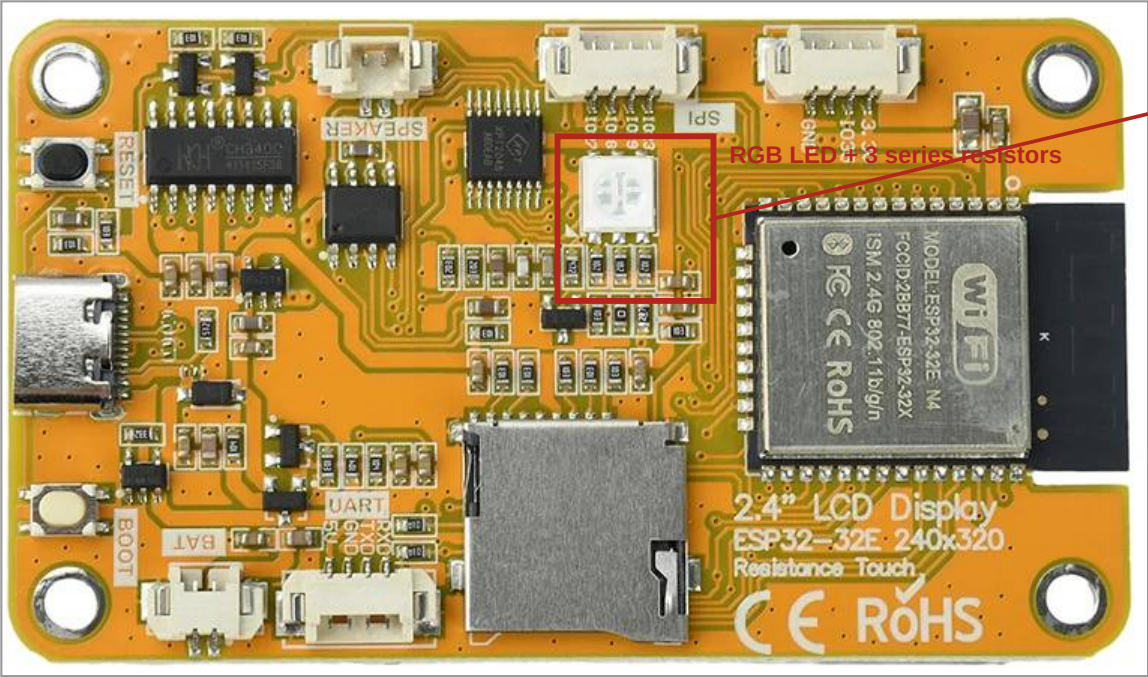


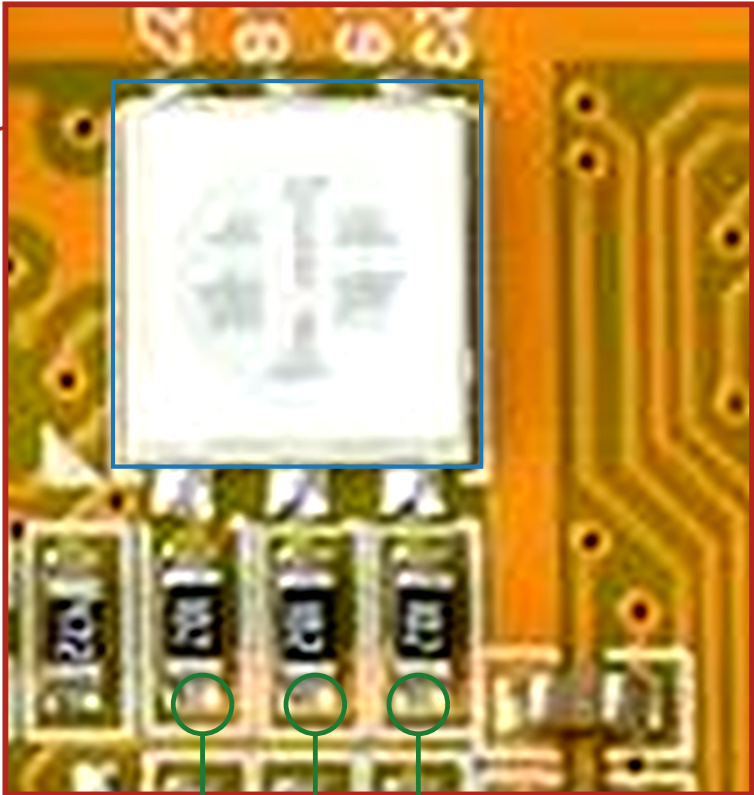
CYD Audio Mod - MAX98357A I2S Amplifier

ESP32-2432S024 (E32R24P) | fallback only - test the onboard amp with tone() on IO26 first | board shown from the BACK



CYD - BACK VIEW

RGB LED (common anode - LOW = on)



SOLDER HERE - outer pad of each resistor (GPIO side)

STEP 1 IDENTIFY (do this first)

I cannot tell which pad is which colour from a photo. Find out with a 5-minute test instead.

Flash this, one pin at a time:

```
void setup(){
  pinMode(16, OUTPUT);
}
void loop(){
  digitalWrite(16, LOW);
  delay(500);
  digitalWrite(16, HIGH);
  delay(500);
}
```

The LED is common anode, so LOW lights it. Note which colour blinks. Repeat for 17 and 22.

Then meter continuity from each resistor's outer pad to that GPIO.

STEP 2 REMOVE THE RESISTOR

Each colour has its own 0603 series resistor sitting directly below the LED. Lift it off with hot air, or walk it off with two irons.

Why remove rather than tack onto it:

Left in place, the LED and resistor hang off your I2S line as a parasitic load. BCLK runs at roughly 1 MHz and LRCLK at 32.8 kHz - the LED will glow faintly, and the extra capacitance is not helping your edges. Removing it fully isolates the LED and leaves a clean, generous pad.

The outer pad (away from the LED) is the GPIO side. That is your solder point.

A 0603 pad is far easier to work with than the LED's own pads or the ESP32-32E castellations. Use 30 AWG silicone wire and strain-relieve with a dot of hot glue or Kapton once it is on.

STEP 3 DISABLE THE ONBOARD AMP

Move the speaker to the MAX98357A.

Unplug it from the CYD's SPEAKER JST - the two amps must not share a driver.

Hold IO4 LOW in firmware.

IO4 is the onboard amp's enable pin (HIGH = on). Drive it low at boot and leave it there, or the muted amp will still draw current and hiss.

```
pinMode(4, OUTPUT);
digitalWrite(4, LOW); // amp off
```

IO26 can then be left alone entirely - audio goes out over I2S instead.

STEP 4 WIRING

MAX98357A	CONNECT TO
LRC / WS	IO16
BCLK	IO17
DIN	IO22
VIN	switched BAT +
GND	BAT -
GAIN	leave floating = 9 dB
SD	board default
+ / -	8 ohm speaker

VIN taps the switched battery line - NOT the 3.3V rail. The amp draws 200-250 mA in bursts. Put the 100-220 uF cap AT the amp pins. I2S pins are assignable in software. SD default = (L+R)/2 mono.

BEFORE YOU REACH FOR THE IRON

- 1. Try the onboard amp first.** Run tone() on IO26 with the speaker plugged into the SPEAKER JST. The Game Boy APU is square waves and noise, which is far more forgiving than music. There is a real chance this sounds fine and the MAX98357A stays in its bag. This whole sheet is a fallback, not a planned step.
- 2. This mod costs you the RGB LED.** IO16 and IO17 are green and blue, IO22 is red. Removing all three resistors kills the LED entirely. It lives inside the shell, so this is no loss - but if you were using it as a boot or charging indicator, move that to the screen first.
- 3. Verify before you cut.** The pin-to-colour mapping above comes from the vendor pin table, and the resistor positions from a photo of the board. Neither is a substitute for the blink test in Step 1. Ten minutes of checking against a board you can actually see beats a lifted pad on unit 7 of 10.
- 4. Practice on a spare.** You are building ten units. Do the first one on whichever board you care least about, and expect the first attempt to be the slowest.